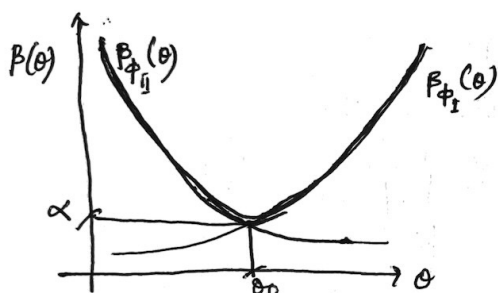


Consider the 2-sided hypothesis $H_0: \theta = \theta_0$ ag. $H_A: \theta \neq \theta_0$.

Claim: UMP (α) test does not exist. Graphically explained.

Let ϕ_I be the UMP(α) test for $H_0: \theta = \theta_0$ ag. $H_1: \theta > \theta_0$

Power for $\phi_I(\theta)$ is increasing in θ
In the region $\theta > \theta_0$, no ^{size α} test has higher power than ϕ_I



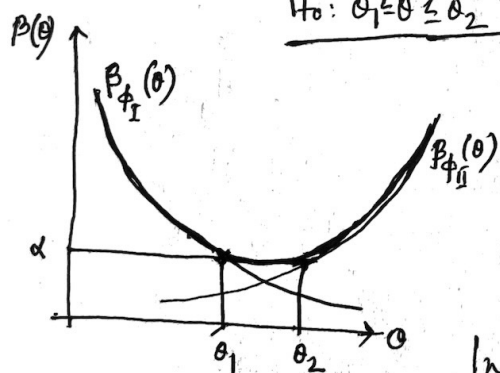
Similarly let ϕ_{II} be the UMP(α) test for $H_0: \theta = \theta_0$ ag. $H_A: \theta < \theta_0$.

Power for $\phi_{II}(\theta)$ is decreasing in θ
In the region $\theta < \theta_0$, no ^{size α} test has higher power than ϕ_{II}

Hence ~~there is~~ a test ϕ which is uniformly most powerful for in both the regions $\theta < \theta_0$ and $\theta > \theta_0$.

Similarly, for the other type of 2-sided test, viz,

$H_0: \theta_1 \leq \theta \leq \theta_2$ ag. $H_A: \theta < \theta_1$ or $\theta > \theta_2$



Let $\phi_I(\theta)$ be the power for $H_A: \theta < \theta_1$

$H_A: \theta > \theta_2$

$\phi_{II}(\theta)$
UMP(α) tests exist for the 2 regions separately,
but not in the combined region as
required by the given H_A .

In both the above situations, power f_n is the convex curve.
This necessitates a third condition in addition to the size & power conditions

Size condition

$$E_{\theta_0} \phi(X) \leq \alpha \text{ or } \sup_{\theta \in \Theta_0} E_{\theta} \phi(X) \leq \alpha$$

Unbiasedness condition
(imposed in above cases)

$$E_{\theta} \phi(X) \geq \alpha \quad \forall \theta \in \Theta_A \quad \left[\begin{array}{l} \text{The power of the test is never} \\ \text{less than the Prob. of Type I error} \end{array} \right]$$

Uniformly Most Powerful condition

$$E_{\theta} \phi(X) \geq E_{\theta} \phi^*(X) \quad \forall \theta \in \Theta_A \text{ \& \> } \phi^* \text{ satisfying the 1st 2 condns}$$

The test ϕ is a Uniformly Most Powerful Unbiased,
UMPU (α) test.

(Not explored any further!)

Likelihood Ratio Tests.

2/29.

(To take care of multiparameter cases & 2-sided hypotheses)

$X_1, X_2, \dots, X_n$ a random sample with pdf/pmf $f_{\underline{\theta}}$,

$$\underline{\theta}_{q \times 1} = (\theta_1, \dots, \theta_q)', \quad \underline{\theta} \in \Theta \subseteq \mathbb{R}^q$$

the likelihood function $L(\underline{\theta}|\underline{x}) = \prod_{i=1}^n f_{\underline{\theta}}(x_i)$

(I) To test $H_0: \underline{\theta} = \underline{\theta}_0$ (fixed) ag. $H_A: \underline{\theta} \neq \underline{\theta}_0$

Define the likelihood ratio criterion,

$$\lambda(\underline{x}) = \frac{L(\underline{\theta}_0|\underline{x})}{\sup_{\underline{\theta} \in \Theta} L(\underline{\theta}|\underline{x})}, \quad \text{clearly, } 0 \leq \lambda(\underline{x}) \leq 1.$$

If the true parameter $\underline{\theta}$ is closer to $\underline{\theta}_0$ (ie, H_0 is the closer to the true situation compared to H_A), $\lambda(\underline{x})$ is expected to be closer to 1.

Further, the true value is from $\underline{\theta}_0$, diff. between $L(\underline{\theta}_0|\underline{x})$ & $\sup_{\underline{\theta} \in \Theta} L(\underline{\theta}|\underline{x})$ increases & $\lambda(\underline{x})$ moves closer to 0.

Hence the critical region is

$$W: \lambda(\underline{x}) < c, \quad \text{where } c \text{ is obtained from the size condition, ie, } P_{H_0}(\lambda(\underline{X}) < c) \leq \alpha.$$

(II) $H_0: \underline{\theta} \in \Theta_0$ ag. $H_A: \underline{\theta} \in \Theta_A = \Theta - \Theta_0$ (ie, unrestricted H_A)

e.g., $\Theta_0 = \{ \underline{\theta}: \theta_j = \theta_{j0} \text{ (fixed)} \forall j=1(1)k; \theta_j \forall j=k+1(1)q \text{ unspecified} \}$
 $\Theta_A = \{ \underline{\theta}: \theta_j \neq \theta_{j0} \forall j=1(1)k; \theta_j \forall j=k+1(1)q \text{ unspecified} \}.$

$$\text{Then the L.R. criterion is } \lambda(\underline{x}) = \frac{\sup_{\underline{\theta} \in \Theta_0} L(\underline{\theta}|\underline{x})}{\sup_{\underline{\theta} \in \Theta} L(\underline{\theta}|\underline{x})} = \frac{\sup_{\theta_{k+1}, \dots, \theta_q} L(\underline{\theta}|\underline{x})}{\sup_{\theta_1, \dots, \theta_q} L(\underline{\theta}|\underline{x})}$$

As before $W: \lambda(\underline{x}) < c$, with appropriate size condition

(III) $H_0: \underline{\theta} \in \Theta_0$ ag. $H_A: \underline{\theta} \in \Theta_A$, $\Theta_0 \cup \Theta_A \subset \Theta$, restricted alternative
 (and, $\Theta_0 \cap \Theta_A = \emptyset$)

$$\lambda(\underline{x}) = \frac{\sup_{\underline{\theta} \in \Theta_0} L(\underline{\theta}|\underline{x})}{\sup_{\underline{\theta} \in \Theta_0 \cup \Theta_A} L(\underline{\theta}|\underline{x})}$$

e.g., in H_A of (II) above replace $\neq$ for the k par. by $>$ or $<$ relation

$W: \lambda(\underline{x}) < c$, & size conditions as before.

In most situations, the exact distn of $\lambda(\underline{x})$ cannot be ascertained.
(Required for computing the cut-off value 'c')

However, for large n , $-2 \log \lambda(\underline{x}) \sim \chi^2_{d.f.}$ with

$$\begin{aligned} \underline{d.f.} &= \dim(\Theta) - \dim(\Theta_0), \quad \dim(\Theta) = \# \text{ unknown parameters in } \Theta \\ &= q - (q-k), \quad \dim(\Theta_0) = k \end{aligned} \quad \Theta_0$$

For example, if $x_1, \dots, x_n$ are iid from $N(\mu, \sigma^2)$, both unknown.

(I) $H_0: \mu = \mu_0, \sigma^2 = \sigma_0^2$ (simple) ag H_A : at least 1 inequality (composite)
Here $\lambda(\underline{x}) \sim \chi^2_2$

(II) $H_0: \mu = \mu_0$ (composite) ag $H_A: \mu \neq \mu_0$ (composite) (unrestricted)
 $\lambda(\underline{x}) \sim \chi^2_1$

(III) $H_0: \mu = \mu_0$ (composite) ag $H_A: \mu > \mu_0$ (composite) (restricted).
 $\lambda(\underline{x}) \sim \chi^2_1$

Ex $x_1, \dots, x_n$ r.s. $N(\mu, \sigma^2)$, $\mu \in \mathbb{R}, \sigma^2 > 0$.

$H_0: \mu = \mu_0$ ag $H_A: \mu \neq \mu_0$

Instead of a UMPU(α) test, consider the LRT(α) test

To be continued ...